

Geometry

Unit 11

Lesson 2 Practice

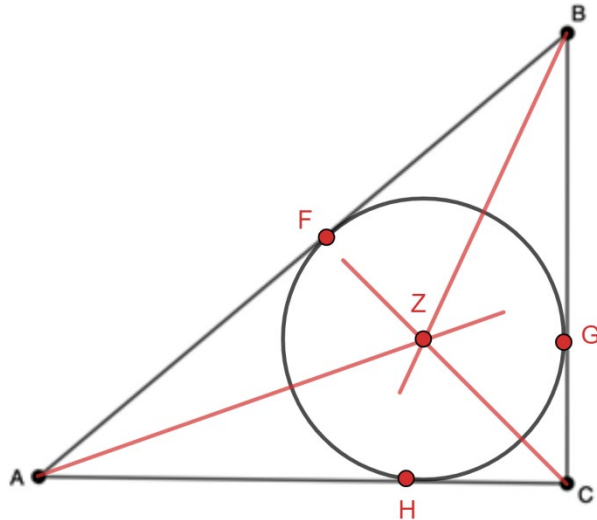
Name **Answer Key** _____

Date _____

Period _____

Use $\triangle ABC$ to answer questions 1-5.

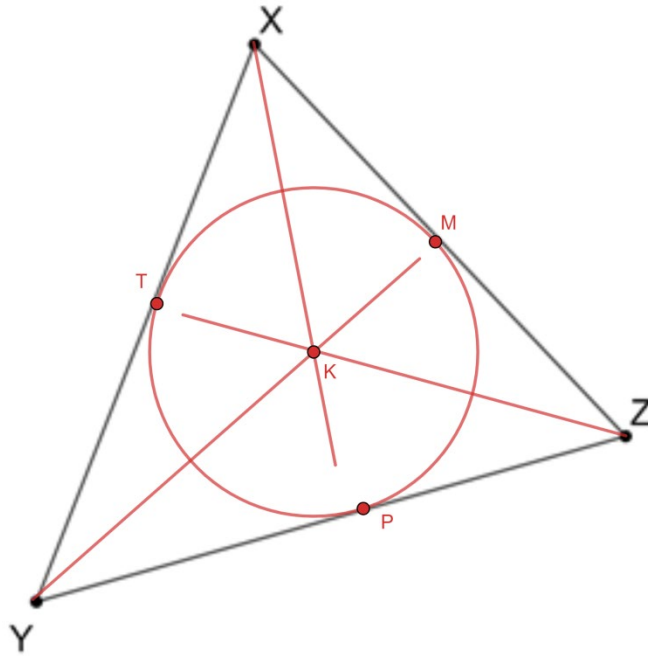
1. Construct the three angle bisectors for $\triangle ABC$.



2. Label the incenter of the triangle as Point Z.
3. Use a compass to draw the inscribed circle inside the triangle, with point Z as its center.
4. Label the three points where the inscribed circle intersects the triangle as F , G , and H .
5. Measure the lengths of segments $\overline{FZ}$, $\overline{GZ}$, and $\overline{HZ}$.
Answers may vary based on unit of measure, where $FZ = GZ = HZ$.

Use $\triangle XYZ$ to answer questions 6-10.

6. Construct the three angle bisectors for $\triangle XYZ$.



7. Label the incenter of the triangle as Point K .
8. Use a compass to draw the inscribed circle inside the triangle, with point K as its center.
9. Label the three points where the inscribed circle intersects the triangle as M , P , and T .
10. Measure the lengths of segments $\overline{MK}$, $\overline{PK}$, and $\overline{TK}$.
Answers may vary based on unit of measure, where $MK = PK = TK$.